Migrating MongoDB to Firestore

Transitioning from Self-Hosted MongoDB to Google Cloud's Managed Serverless Infrastructure with MongoDB Compatibility.

Native vs. MongoDB Compatibility

Firestore Native

Proprietary SDKs optimized for real-time mobile/web sync and offline support. Best for new Google-native ecosystems.

Challenge: Requires code rewrites and proprietary query language adoption.

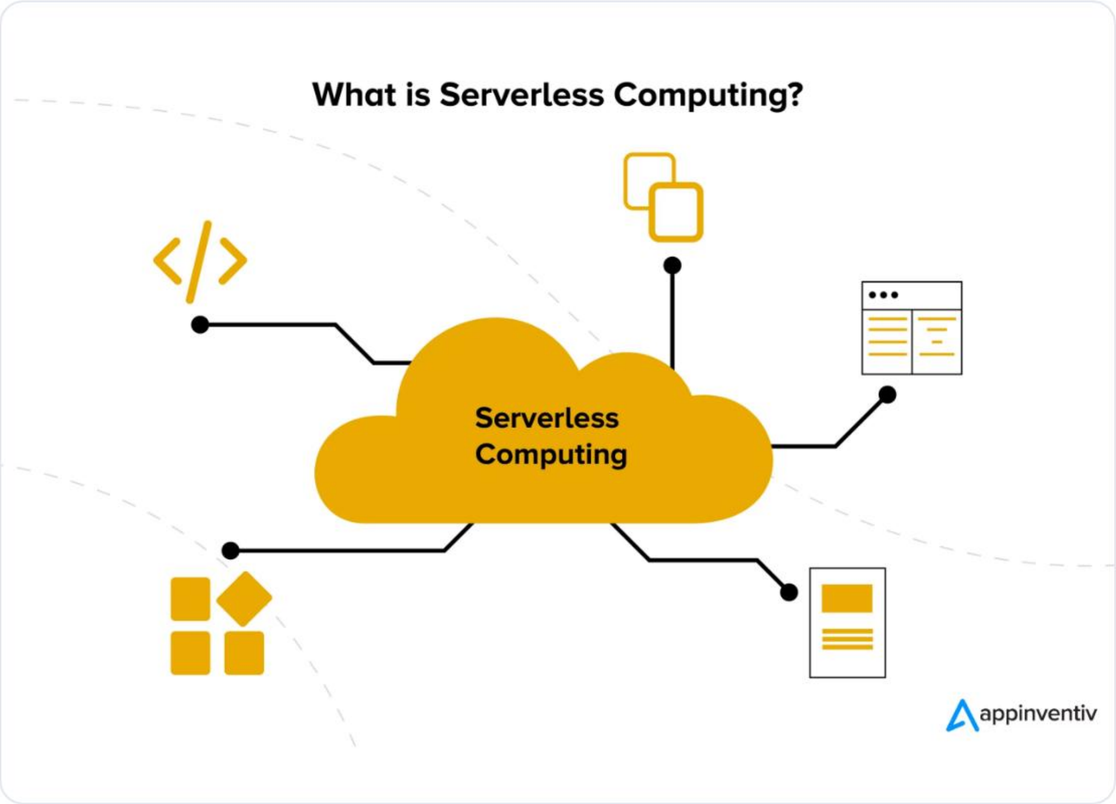
MongoDB Compatibility

API layer allowing existing MongoDB drivers, tools (Compass), and queries to work on Firestore with minimal changes.

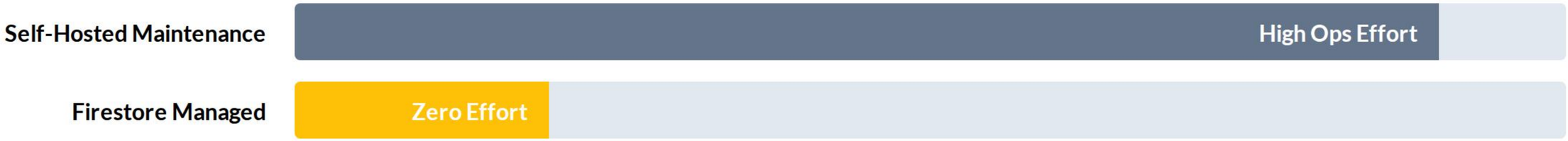
Benefit: Retain your code while gaining serverless scalability and SLA.

Management & Scaling Benefits

- **Zero Infrastructure:** Google manages sharding, replication, and patching.
- **Automatic Scaling:** Effortlessly handle traffic spikes without manual intervention.
- **High Availability:** Up to 99.999% SLA with multi-region consistency.



Reliability and Cost Efficiency



"Switching to serverless shifts focus from administration to core feature development."

Strategy for Empty Databases

For brand-new projects, migration is a simple 3-step configuration change:

1. Create Firestore in MongoDB compatibility mode via GCP Console.
2. Extract the unique MongoDB Connection String (URI).
3. Update your app configuration and reconnect instantly.

Console View

Moving Existing Data Streams

Google Datastream

Best for large-scale, near-zero downtime needs. Uses CDC and Dataflow for bulk loading.

Custom Batch Job

Ideal for < 500GB. Python/Spring Boot script with BulkOperations for full control.

Phased Migration

Read from both while writing to new Firestore to ensure zero data loss.

Post-Migration Architecture

App Layer: Standard Mongo drivers.

Storage: Google Global Infrastructure.

Security: IAM access control.

Analytics: Easy export to BigQuery.

Field and Collection Restrictions

Constraint	Restriction Detail	Action Required
Naming	Cannot start with __ or contain $	Rename invalid fields
System Collections	No `system.*` collections	Validate schema
_id Type	Limited to specific BSON types	Convert unsupported types

Document Size & Nesting Limits

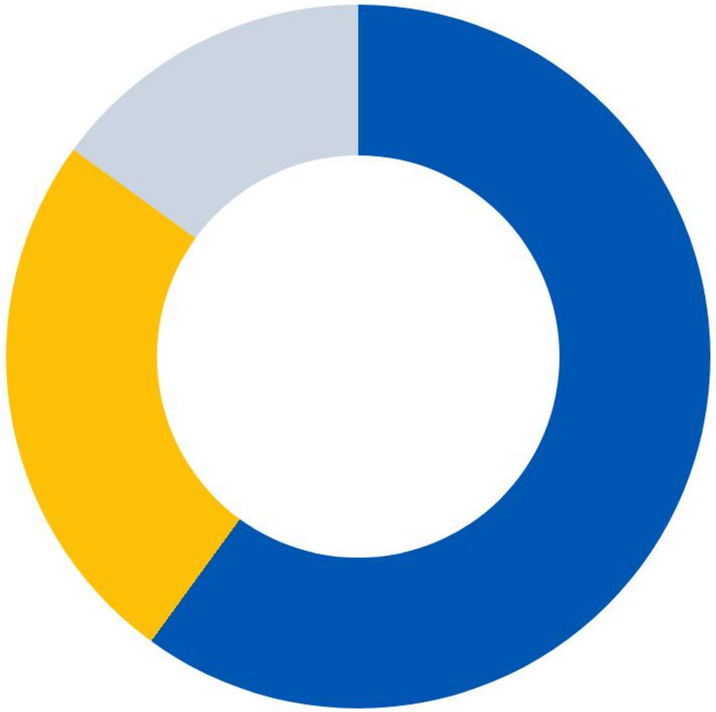
4 MiB

Max Document Size
(Vs 16 MiB in standard MongoDB)

20

Max Nesting Levels
Fields, Objects, and Arrays

Operational Cost Breakdown



- **Data Operations (60%):** Read/Write ops.
- **Storage (25%):** Indexed storage volume.
- **Other (15%):** PITR, backups, networking.

Recommendations & Next Steps

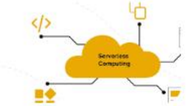
- **Run Pre-migration Scans:** Identify documents > 4MB and invalid field names.
- **Proof of Concept:** Migrate a non-critical collection using a Python batch job.
- **Choose Configuration:** Select Regional vs Multi-Region based on availability needs.
- **Monitor:** Use Google Cloud Billing tools to optimize costs after launch.

Questions?

Thank you for your attention.

Cloud Migration Team | Solution Architecture

Image Sources



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